Mathematical Optimization

Types of Optimization

Discrete/Continuous Optimization

If the variables can only take specific discrete values, the optimization problem is considered discrete; otherwise, it is continuous.

For discrete optimization, common types include:

- **Combinatorial Optimization**: Optimization problems where the solution space is discrete and finite, often involving arrangements or selections of objects.
- **Integer Programming**: Optimization problems where some or all of the variables are restricted to be integers.

Unconstrained/Constrained Optimization

If there are restrictions or constraints on the variables, the optimization problem is considered constrained; otherwise, it is unconstrained.

Linear/Nonlinear Programming

If the objective function and constraints are linear, it is linear programming; otherwise, it is nonlinear programming.

Optimization Algorithms

Global/Local Optimization

If x^* is the best solution in the entire solution space, it is a global optimum; if it is only the best in a small neighborhood, it is a local optimum.

- First-order necessary condition for local optimum: $\nabla f(x^*) = 0$
- Second-order necessary condition for local optimum: $\nabla^2 f(x^*)$ is positive semi-definite.
- Second-order sufficient condition for local optimum: $\nabla^2 f(x^*)$ is positive definite.

Gradient Descent Method

For $\min f(x)$, the update rule is:

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k)$$

where α_k is the step size.

Newton's Method

For $f(x) = 0$, the update rule is:

$$x_{k+1} = x_k - [\nabla f(x_k)]^{-1} f(x_k)$$

For $\min f(x)$, the update rule is:

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$$

Lagrange Multipliers

For constrained optimization problems of the form:

$$\min f(x)$$

subject to

$$g_i(x) = 0, \quad i = 1, \ldots, m$$

the Lagrangian is defined as:

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x)$$

The necessary conditions for optimality are:

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0$$

$$g_i(x^*) = 0, \quad i = 1, \ldots, m$$

Karush-Kuhn-Tucker (KKT) Conditions

For optimization problems with inequality constraints:

$$\min f(x)$$

subject to

$$g_i(x) \leq 0, \quad i = 1, \ldots, m$$

$$h_j(x) = 0, \quad j = 1, \ldots, p$$

the KKT conditions are:

1. Stationarity:

$$\nabla f(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla g_i(x^*) + \sum_{j=1}^{p} \mu_j^* \nabla h_j(x^*) = 0$$

2. Primal feasibility:

$$g_i(x^*) \leq 0, \quad i = 1, \ldots, m$$

$$h_j(x^*) = 0, \quad j = 1, \ldots, p$$

3. Dual feasibility:

$$\lambda_i^* \geq 0, \quad i = 1, \ldots, m$$

4. Complementary slackness:

$$\lambda_i^* g_i(x^*) = 0, \quad i = 1, \ldots, m$$